# A family of quartics in 301 variables

9 November 2026

**Abstract**

We specify a family of integer quartics in 301 total variables by 275 quadratic equations and complete coefficient tables. Under the arithmetic premises of the elliptic source construction, every sufficiently large member defines the integers in the rationals. Two height detectors use the rational trace of a sum of elliptic abscissas. The sixteen local tests are represented by polynomial expressions with distinct valuation classes, using fourth and eighth powers already computed for the formal logarithm. The coefficient family is explicit, but the sufficient parameter threshold is ineffective because the aggregate-trace height theorem has an ineffective additive constant. No particular defining member of this family is selected. A fully specified 351-variable quartic is a separate result.

## 1 The exact family and its scope

All variable counts include the input $X$. Put

$$A_N = 80989^{2N+1} \qquad (N \geq 1).$$

The tables `quadratic-family301.json` and `polynomial-family301.json` specify 275 primitive integer quadratic rows $Q_{N,j}$ in 301 coordinates and

$$P_N = \sum_{j=1}^{275} Q_{N,j}^2.$$

Neither $A_N$ nor $N$ is a variable of $P_N$.

**Theorem 1.1.** *Under the arithmetic premises in Section 2, there is an integer $N_0 \geq 27464$ such that, for every $N \geq N_0$ and $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{300},\ P_N(x, \boldsymbol{y}) = 0.$$

*For every $N \geq 1$, $P_N$ has degree exactly four, content one, and coefficient height*

$$H(P_N) < 2^{1520} 80989^{4N+2}.$$

*For $N \geq 27464$ it has 4305 nonzero coordinate monomials. The proof does not determine a numerical value of $N_0$.*

An exact family specification and an effective choice of a defining member are different assertions. The theorem supplies the first together with eventual definition of $\mathbb{Z}$; its height argument does not supply the second. The explicit 351-variable theorem in [2] uses three height detectors and an effective sum-height inequality.

## 2 Arithmetic premises

Let $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, $d = 3 - \rho$, and

$$E : y^2 = x^3 - d^2x, \qquad P = 2(1 + \rho, 2\rho) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32}\right).$$

The arithmetic input contract is the one precisely stated in [4, Section 2] and restated in [2, Section 2]:

1. $2E(K) = \mathbb{Z}P$ and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer $n$.
2. The five-contact height bound $h(t) \leq \mathcal{H} M_5(t)^{19}$, with

$$\mathcal{H} = 2^{110}210^{1800}\left(\prod_{p \leq 76176} p\right)^{19} + 128 < 2^{2093448}.$$

   Here $M_5(u/v)$ is the product of primes greater than 76176 with positive odd valuation on at least one of $v, u, u - v, u - 2401v, 32u + 343v$, in lowest terms with $v > 0$, and the excluded contact values are $0, 1, 2401, -343/32$.
3. The source pole obstruction outside $2, 3, 17$ and its simultaneous realization for inputs in $(0, 17)$ with square reduced denominator and the specified local squareclasses and valuation bounds at $2, 3, 17$. The normalized seven-coordinate equations are retained literally.
4. The simultaneous prime-pattern and conjugate reduction-order conditions with arbitrary fixed odd avoidance modulus, finite congruences, and real approximation, including the quantitative weighted selection premise stated in the cited contract. In particular the slowly decreasing separation parameter is available after all fixed data are chosen.

No finite coefficient computation proves these inputs. The aggregate-trace height theorem used next is a proved consequence of Von Buhren's family height theorem, with an ineffective additive constant, as detailed in [5]. It is not an additional conjecture.

## 3 The trace height and the two local detectors

The retained square charts give root points

$$Q_1 = [n]P, \quad Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P, \qquad B = \max(|n|, |a|, |t|, |b|),$$

with nonzero integer labels. Their interpretation for all nonzero integer labels, without a parity restriction, is proved in [3, Section 2]. Define

$$S_0 = \frac{1}{2}\operatorname{Tr}_{K/\mathbb{Q}}\left(\sum_{i=1}^{4} x(Q_i)\right).$$

There are fixed constants $M_1 > 0, M_2 \geq 1$ such that

$$B^2 \leq M_1 h(S_0) + M_2. \tag{1}$$

The additive constant is not effective in the cited proof. Every root abscissa is positive at both real embeddings, so $S_0 > 12$.

Let $\sigma$ denote conjugation on $K/\mathbb{Q}$ and let $f = 1/x$, extended by zero at the identity. The auxiliary comparison points satisfy $U_0 = H$, $V_0 = H - P$, and $Q_i = U_i - V_i$ as group differences on $E$. These are the exact addition relations of the retained point charts in [3].

Use the two nonzero rational square scales $1, S_0^2$. For each scale $a_*$ put $t_* = a_*\lambda$, $\delta = t_* - 1$ and

$$R_* = \frac{\prod_{\theta\in\Theta}(t_* - \theta)}{a_*^5\delta^6}, \qquad q_m = R_*^{1-2m}\delta^{-2m-2}, \qquad \Theta = \{0, -1/2, -1, 1/2399, -32/407\}.$$

These formulas are used on their nonzero domain; the exact system consists of their cleared quadratic equations. The case $t_* = 1$ has empty contact radical and requires no reciprocal.

Define $A_i = f(U_i)$, $B_i = \sigma f(V_i)$, and

$$\begin{aligned} C_i &= L_i - L_i' \quad (1 \le i \le 4), & I &= L_1 - \beta, \\ D_H &= f(H), & G &= L_3^2 - L_2(L_2 - 2)L_3L_4 + (L_2 - 2)^2L_4^2 - 1. \end{aligned}$$

The alternate ratios $L_i'$ already include conjugation. The nine tests $A_i, B_i$ have order one, $C_1, I$ have order ten, and $D_H, C_2, C_3, C_4, G$ have order forty. For an ordered list of tests, use the factor

$$\mathcal{T}(D_0, \dots, D_{s-1}) = 1 + \sum_{j=0}^{s-1} q_{m(D_j)}^{2^j} D_j^{2^{j+1}},$$

where $m(D_j)$ is its stated order. Thus different required orders can occur within one factor.

The detector uses the product of the factors for the following six lists:

$$\begin{gathered} (D_H, C_1, B_0), \quad (C_2, I, B_2), \quad (C_3, A_2, B_3), \\ (C_4, A_3, B_4), \quad (G, A_4), \quad (A_1, B_1). \end{gathered}$$

Each of the sixteen tests occurs exactly once. The common-place valuation argument and the literal cleared trace equations are proved in [1]; each contact prime supplies every required test order at one place. The rational logarithm approximants and the scalar point exclusions are retained.

If a nonzero integer $J$ is divisible by the $k$th powers of both contact radicals, the contact-height theorem bounds $h(\lambda)$ and $h(S_0^2\lambda)$ by $(\mathcal{H} + \log 3)|J|^{19/k}$. Since $\lambda \neq 0$,

$$2h(S_0) = h(S_0^2) \le h(S_0^2\lambda) + h(\lambda).$$

Combining this with (1) gives

$$B^2 \le C|J|^{19/k} + M_2, \qquad C = M_1(\mathcal{H} + \log 3). \tag{2}$$

## 4 Soundness for every sufficiently large parameter

Put $\beta = 51X^2 + 2504$ and $c = |a - 1|$. The retained scalar circuit uses the positive contact product

$$r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1)$$

and normalizer $R_s = A_N r/(1 + 80989^2 r)$, with shift $R_s + 9792$. The fixed-prime argument gives

$$c \ge 80989^{L_N}, \quad L_N = \left\lfloor \frac{187(2N - 1)}{8} \right\rfloor + 1, \quad h(\beta) \le \mathcal{H}c^{19/24}. \tag{3}$$

Its integral group identities and exceptional-point exclusions are unaffected by the local test factors.

*Proof of soundness in Theorem 1.1.* Set

$$\Delta = t^2 - a(a - 2)tb + (a - 2)^2b^2 - 1.$$

The formal comparison gives divisibility of $\Delta$ by the fortieth powers of both contact radicals. If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$, and (2) gives

$$B^2 \leq C14^{19/40}B^{19/10} + M_2.$$

This is impossible for sufficiently large $B$. Equation (3) ensures $B \geq c - 1 \to \infty$ uniformly as $N \to \infty$. Hence $\Delta = 0$ for every sufficiently large $N$.

The integral Lucas descent in [4] then gives $\log B > |a| - 3 \geq c - 4$. Suppose $n - \beta \neq 0$, and let $J$ be its nonzero reduced numerator. The order-ten tests imply that both tenth powers of the contact radicals divide $J$, while $|J| \leq 2B\exp(h(\beta))$. For $B^2 \geq 2M_2$, equation (2) therefore implies

$$\log B \leq 10\log(2C) + 19\log 2 + 19h(\beta) \leq 10\log(2C) + 19\log 2 + 19\mathcal{H}c^{19/24}.$$

The right side is $o(c)$, contradicting the Lucas lower bound for sufficiently large $c$. Thus $n = \beta \in \mathbb{Z}$. Writing $X = u/v$ in lowest terms, $v > 0$, gives $v^2 \mid 51$ and hence $v = 1$.

All thresholds depend only on the fixed constants $C, M_2, \mathcal{H}$. Since $L_N$ increases, one integer $N_0 \geq 27464$ works for every $N \geq N_0$. The dependence on $M_2$ is precisely the ineffective part of the assertion. □

## 5 Completeness and exact verification

Fix an integer input and a positive integer $N$. The scalar preparation and the Lucas construction select the same integer root labels and nonzero root points as in [3]. Fixed real neighborhoods, finite preparation depths, and an avoidance modulus are chosen before the contact primes. The modified factors introduce no new rational function poles and retain the same formal-depth requirements.

The simultaneous completeness proposition of [1] applies to both scales: at every selected contact prime the cleared trace has a unit value congruent to 1 or 2; the small-prime preparation is preserved by taking powers of its already small terms; and the real factors tend uniformly to 1 as $\lambda \to +\infty$. Fresh pole witnesses therefore exist under exactly the retained arithmetic contract. Filling the rational arithmetic circuit gives all 275 quadratic rows. Their sum of squares vanishes, proving completeness and Theorem 1.1.

All required fourth and eighth powers of the $B_i$ are explicit linear forms in existing logarithm coordinates. The five new field powers are $C_1^4, I^4, A_2^4, A_3^4, A_4^4$, each computed by squaring its existing square. Since every order-forty test occupies position zero, no square or cube of $q_{40}$ is retained. The complete accounting and both directions of the local argument are in [1]. The resulting system has 301 coordinates and 275 quadratic rows.

Run the following commands from `computation/quartic`:

```
python build_quartic_mixed.py
python check_quartic_mixed.py
```

The independent checker verifies every coefficient of the quadratic system and the quartic. It confirms 4320 terms when powers of the fixed parameter are recorded separately, and 4305 ordinary monomials for $N \geq 27464$. The coefficient-height bound and a retained coefficient equal to one give the asserted height and content. The checks do not select $N_0$ or prove the arithmetic premises.

## References


[1] *Simultaneous valuation conditions and explicit quartics*, 9 November 2026, accompanying local theorem, simultaneous completeness proof, exact circuit, and independent verification. PDF; source: `mixed_valuation_tests_20261109_v11.tex`.

[2] *An explicit quartic in 351 variables*, 9 November 2026, specified polynomial and effective conditional projection theorem. PDF; source: `quartic351_20261109_v11.tex`.

[3] *A family of quartics in 317 variables from aggregate trace heights*, 8 November 2026, full point-chart, trace-height, and scalar arguments. PDF; source: `quartic317_family_20261108_v9.tex`.

[4] *A forced prime condition and an explicit quartic in 392 variables*, 8 November 2026, arithmetic input contract and scalar/Lucas arguments. PDF; source: `quartic392_20261108_v8.tex`.

[5] *Aggregate trace heights of elliptic multiples*, 8 November 2026, proof of the linear height comparison from Von Buhren's family height theorem. PDF; source: `aggregate_trace_height_20261108_v9.tex`.

[6] E. Glazer, *An explicit quartic defining the integers in the rationals*, Version 13, manuscript and coefficient data, commit `5bbd74f6a479776a332cbe036809d5dddae63720`, https://github.com/elliotglazer/diophantine-z-q.